

Design and Simulation of an Automatic Street Light Control System

DDCA Project Presentation

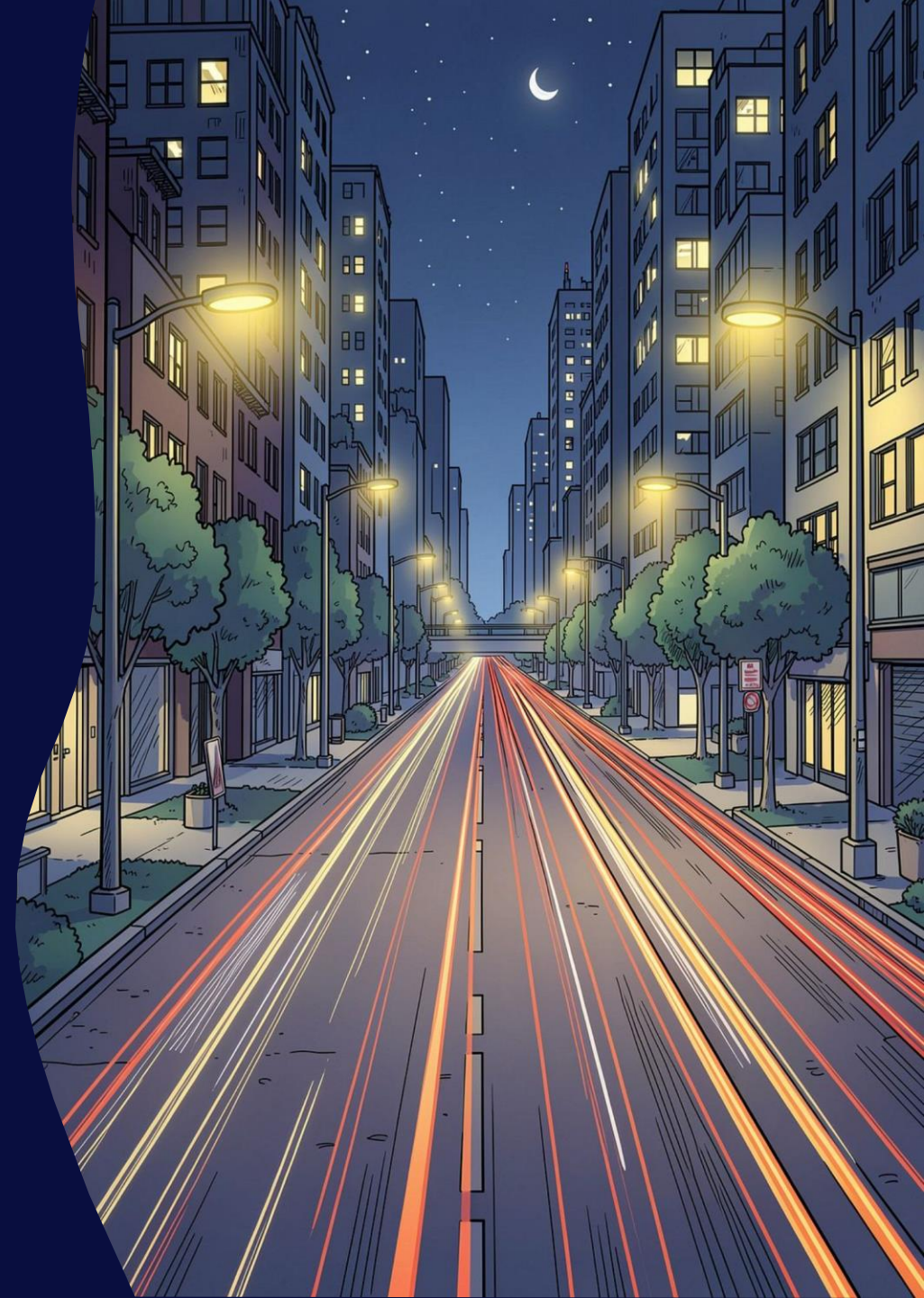
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Introduction: Smart Street Lighting for a Brighter Future

Our project introduces an advanced Automatic Street Light Control System, designed to revolutionize urban illumination. By integrating sensor technology and intelligent algorithms, this system aims to optimize energy consumption, enhance public safety, and reduce operational costs.

We delve into the detailed design and simulation of a solution that dynamically adjusts lighting levels based on real-time environmental conditions, paving the way for more sustainable and efficient city infrastructure.



The Problem: Energy Waste in Street Lighting

Always-On Lighting

Manual control and fixed timers keep lights on at full brightness even on empty roads.

High Energy Demand

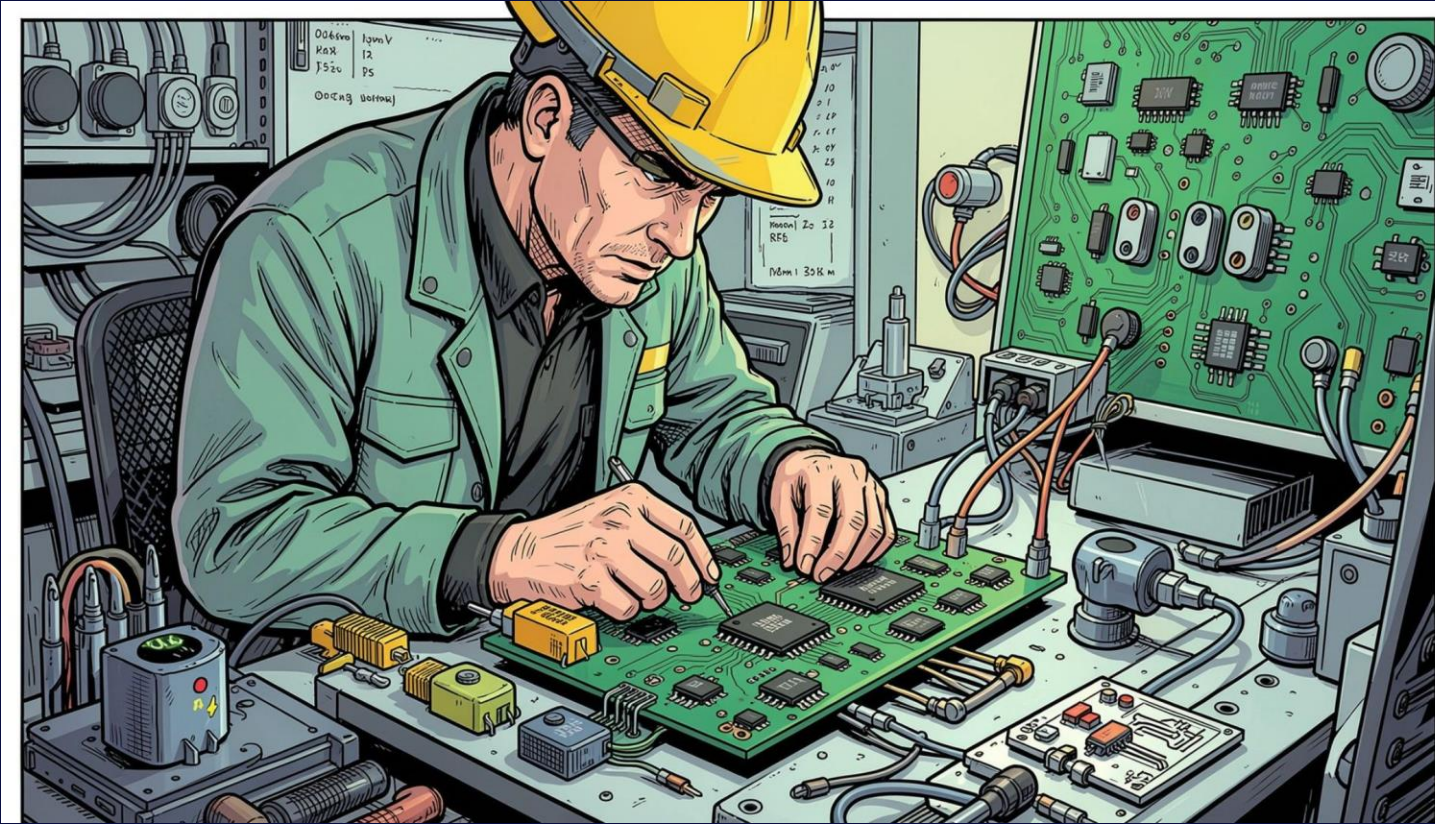
Conventional street lighting consumes a major share of a city's total electrical energy budget.

Human Error & Seasonal Drift

Manual operation is susceptible to human error and fails to adapt to seasonal timing variations.



Project Objectives



1 Automate ON/OFF Control

Use ambient light sensing (LDR) to switch street lights automatically based on day/night conditions.

2 Dynamic Brightness Adjustment

Detect motion via IR/PIR sensors and adjust lamp brightness in real time for safety and efficiency.

3 Reduce Consumption & Extend Lifespan

Lower overall energy use and reduce wear on LED lamps through intelligent dimming and switching.

Literature Survey / Existing System

Automatic Street Light Control System

1 Manual Control



- Human-operated ON/OFF
- Requires manpower
- Energy waste

2 Timer-Based Control



- Fixed schedules
- Cannot respond to changing daylight or traffic

3 LDR-Based Control



- LDR senses ambient light
- OFF in daytime
- ON at night

4 LDR + PIR/IR + Arduino



- Darkness + movement
- Adaptive bright/dim operation
- Saves more energy

5 IoT / Smart Lighting



- Remote monitoring
- Fault detection
- Adaptive control
- Energy optimization



Research Gap: basic systems mainly switch lights using light level or fixed timing; newer systems combine sensing, adaptive dimming, monitoring and fault detection.

Selected literature



IEEE 2021:
Arduino + LDR + IR + RTC
for automatic intensity regulation.



Sensors/MDPI 2018:
Arduino + LDR + IR,
dim at night and brighten
on object detection.



IET Smart Cities 2020:
adaptive dimming for
energy and carbon reduction.



2025 studies:
automated lighting with LDR/PIR
and fault detection.

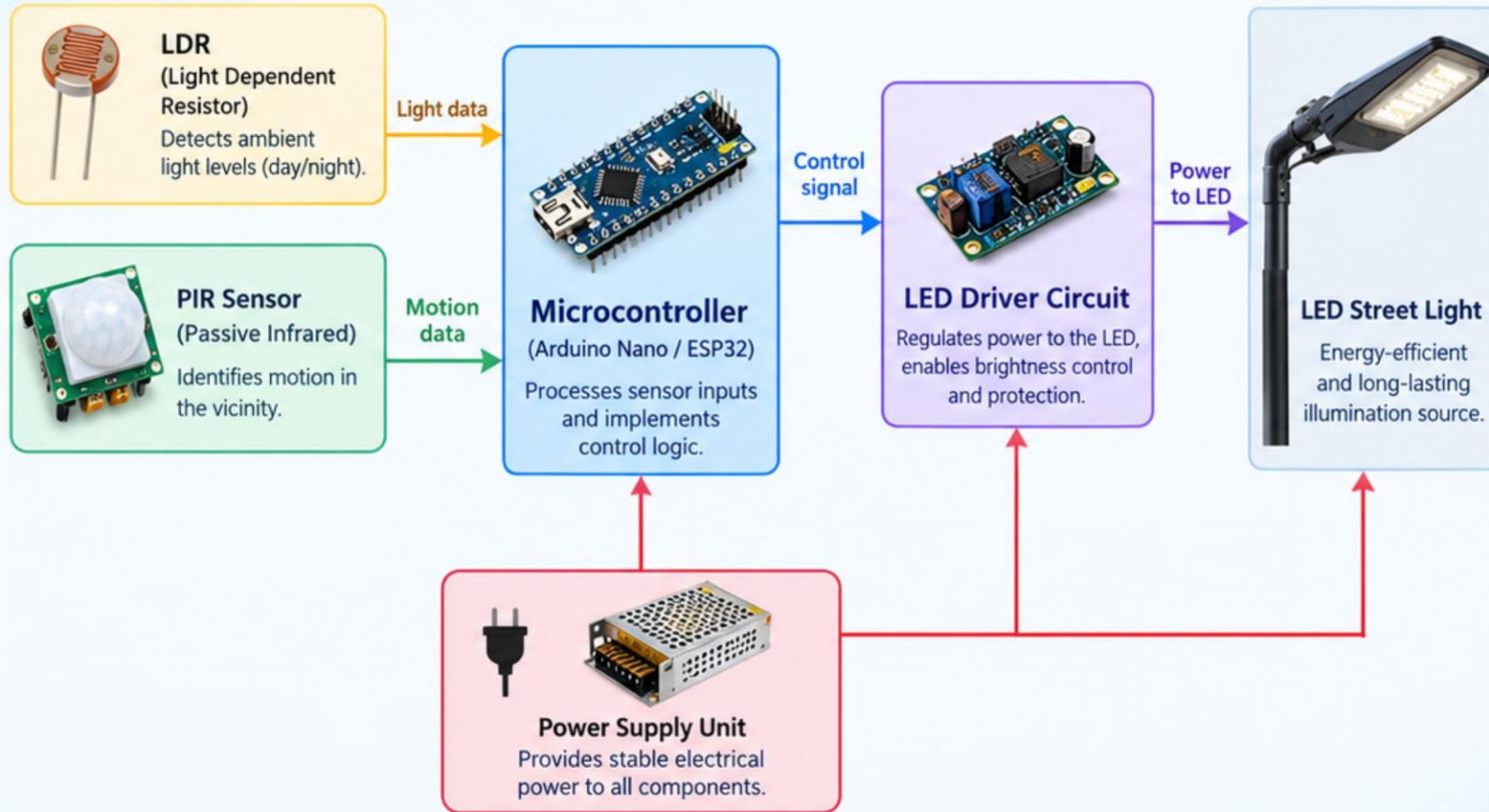


System Architecture & Hardware Components

Our automatic street light system integrates several key components to achieve intelligent and energy-efficient operation, dynamically responding to environmental conditions.

Smart Light
for a Brighter
Tomorrow

System Architecture



Key Hardware Components

-  **1 Microcontroller**
Processes sensor inputs and implements control logic (e.g., Arduino Nano, ESP32).
-  **2 Light Dependent Resistor (LDR)**
Detects ambient light levels to determine day or night conditions.
-  **3 Passive Infrared (PIR) Sensor**
Identifies motion in the street light's vicinity for dynamic brightness adjustment.
-  **4 LED Street Light**
The illumination source, chosen for its energy efficiency and longevity.
-  **5 LED Driver Circuit**
Regulates power to the LED light, enabling brightness control and protection.
-  **6 Power Supply Unit**
Provides stable electrical power to all system components.



Energy Efficient



Reliable Performance



Modular Design



Easy Maintenance

Working Principle



Daytime

LDR detects sunlight → lamp stays **OFF**, conserving energy throughout daylight hours.



Night — No Motion

Lamp operates at a **dim level** for baseline visibility and significant energy savings.



Night — Motion Detected

IR/PIR triggers lamp to switch to **full brightness** immediately for pedestrian and vehicle safety.

Control Logic & Implementation

Basic Approach

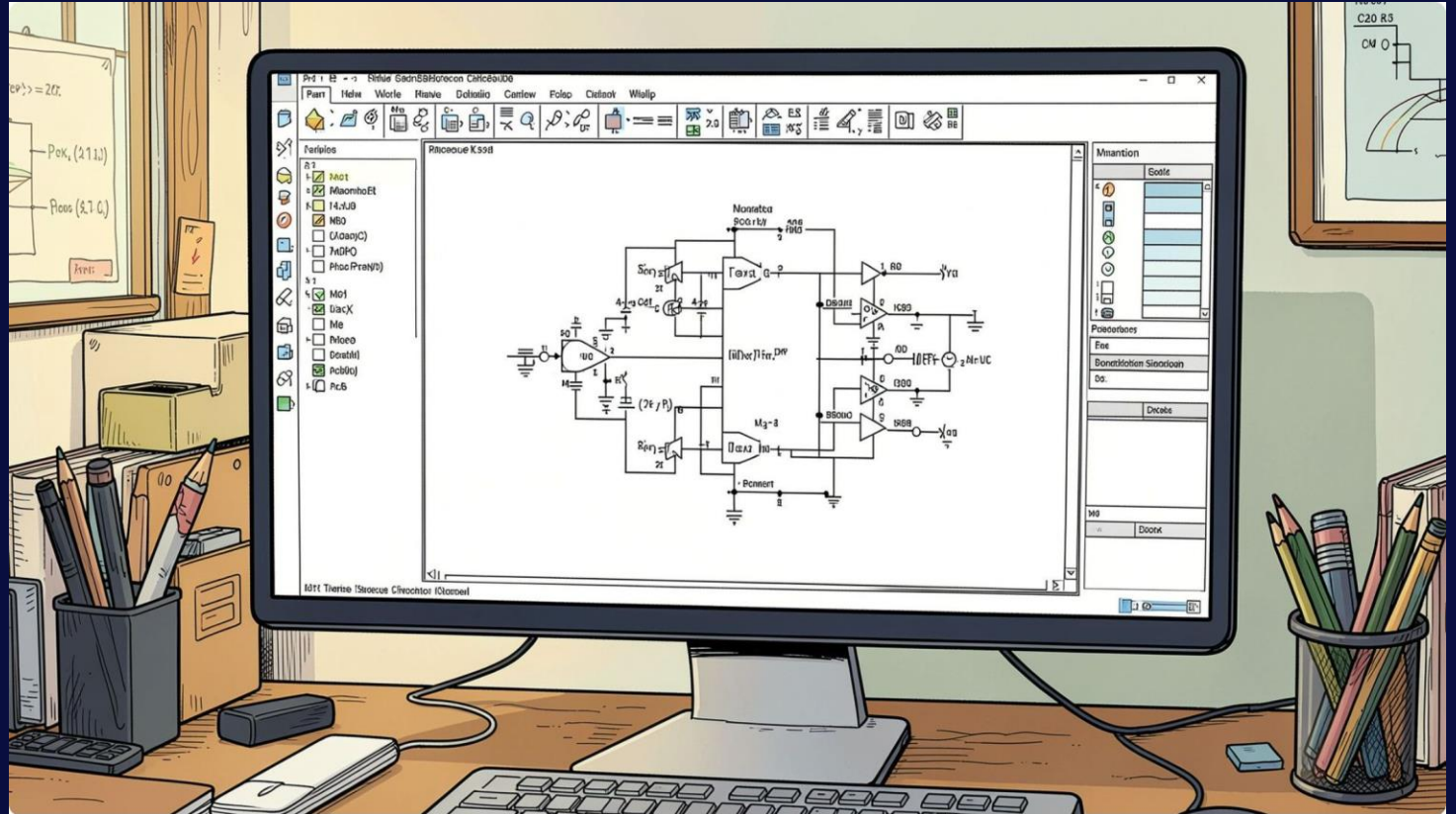
A transistor circuit operates in **saturation (ON)** and **cut-off (OFF)** regions to switch the lamp based on sensor input — a straightforward digital logic implementation.

Advanced Approach

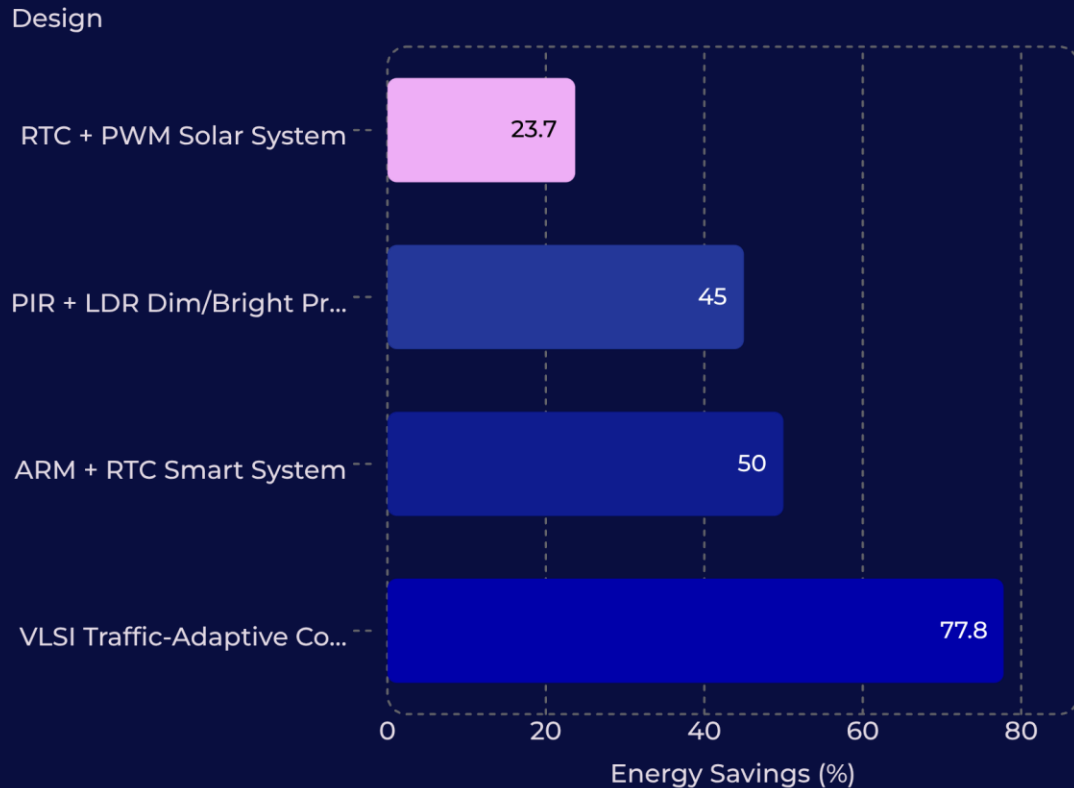
A microcontroller divides the night into time slots and applies **Pulse Width Modulation (PWM)** for variable brightness control, enabling fine-grained energy management.

Simulation Tools

- **Proteus 8** — circuit-level simulation and validation
- **Quartus & Verilog HDL** — FPGA-based digital design and synthesis



Results: Energy Savings Across Designs

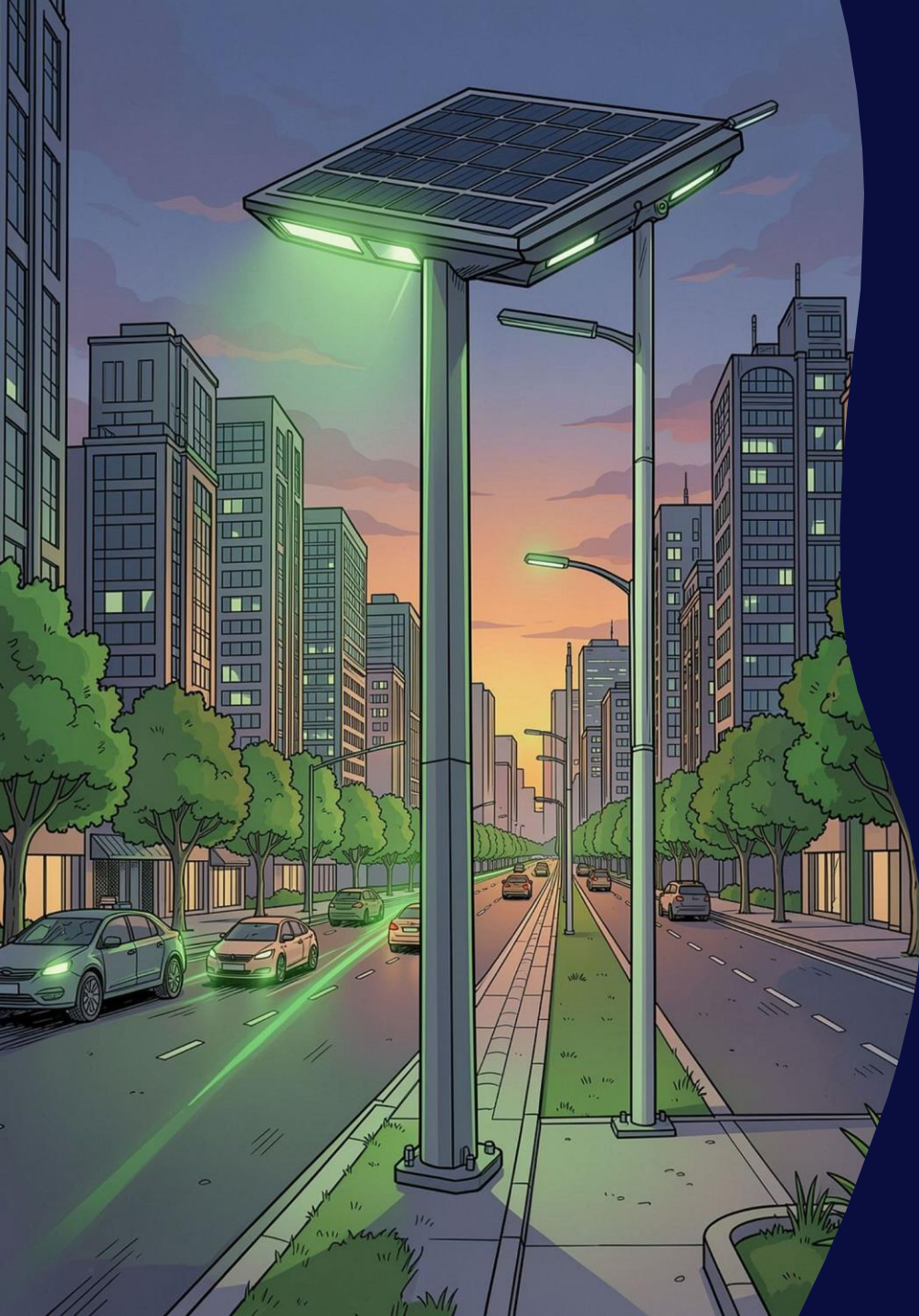


Key Findings

Across all reviewed designs, intelligent street light controllers consistently outperform conventional fixed-timer systems:

- **RTC + PWM solar system:** 23.7% energy savings
- **PIR + LDR dim/bright prototype:** ~45% estimated reduction
- **ARM + RTC smart system:** 50% excess power reduction
- **VLSI traffic-adaptive controller:** 77.78% power reduction

✓ More sophisticated sensing and control logic yields dramatically greater energy savings.



Key Advantages & Impact



Energy Efficiency

Avoids unnecessary full-night lighting by dimming when roads are empty, cutting consumption significantly.



Improved Safety

Instantly brightens when people or vehicles are detected, ensuring visibility exactly when it matters.



Lower Maintenance

Automation and fault monitoring reduce manual intervention and extend LED lamp lifespan.



Renewable Integration

Supports solar-powered operation, enabling fully off-grid and sus

Conclusion

Our project successfully designs and simulates an automatic street light control system that addresses significant energy waste in conventional setups. By dynamically adjusting lighting based on real-time ambient conditions and motion detection, the system achieves substantial energy savings, enhances public safety, and reduces operational costs.

This intelligent solution represents a crucial step towards creating smarter, more sustainable urban infrastructures, contributing to a greener future with improved efficiency and responsiveness.

